PROJECTSHOWCASE CHALLENGE — MINDSPARKS 2026 · AUSTIDC

Project Abstract — UniGPT

An AI Academic Copilot for Smart University Management

---

1. Problem Statement

Modern universities still rely heavily on fragmented systems and manual workflows for managing academic activities. Students frequently struggle to access important course materials, notices, assignment guidelines, attendance records, and academic policies because these resources are scattered across multiple disconnected platforms such as ERP systems, WhatsApp or Messenger groups, Google Classroom, spreadsheets, and paper-based records.

Faculty members face similar challenges, spending significant time on repetitive tasks such as attendance tracking, assessment creation, grading, feedback generation, and academic communication. Administrative departments also lack real-time institutional visibility, making decision-making slow and inefficient.

2. Innovation & Uniqueness

UniGPT addresses this problem by introducing an intelligent, centralized academic platform powered by Artificial Intelligence. Rather than forcing students, faculty, and administrators to navigate multiple disconnected tools, UniGPT unifies all major academic workflows into a single platform with role-based experiences. The system provides one integrated ecosystem for academic communication, course management, attendance, assessments, document sharing, analytics, and AI-assisted decision support.

The key innovation of UniGPT lies in its Retrieval-Augmented Generation (RAG) architecture. Unlike generic AI chatbots such as ChatGPT that may generate inaccurate or hallucinated responses, UniGPT grounds every AI response in institution-approved documents such as syllabi, policies, handbooks, lecture materials, and academic resources.

When a user asks a question, the system retrieves relevant document passages, performs contextual reasoning, and generates a precise answer with source citations and confidence scores. This ensures that responses are trustworthy, auditable, and aligned with institutional rules.

3. AI Usage

UniGPT uses advanced Artificial Intelligence through **Large Language Models (LLMs)** and **Retrieval-Augmented Generation (RAG)** to deliver accurate academic assistance. The platform integrates **OpenAI models**, specifically **GPT-4o** for reasoning and response generation, and **text-embedding-3-large** for semantic document retrieval.

When a user asks a question, UniGPT converts both the query and institution-approved academic documents into vector embeddings. It then performs semantic similarity search to retrieve the most relevant document chunks, such as syllabus pages, policies, lecture materials, or handbooks. These retrieved passages are provided as context to the language model, allowing the AI to generate responses grounded in institutional knowledge rather than generic internet data.

AI powers multiple intelligent workflows inside UniGPT, including:

- Grounded academic Q&A with source citations, confidence scores, and suggested follow-up questions
- Multiple AI chat modes including academic support, exam preparation, research, assignment help, and career guidance
- Automated quiz, assignment, and class test generation, allowing faculty to create assessments manually or with AI assistance
- AI-assisted grading feedback to help faculty evaluate submissions faster and more consistently
- Personalized student guidance based on academic context and progress
- Operational analytics and early risk detection for identifying engagement or performance issues

4. Core Features

UniGPT offers comprehensive core features tailored to three major user groups.

Students receive an AI academic copilot that helps them access course materials, track attendance, monitor GPA/CGPA, manage deadlines, and receive instant academic assistance.

Faculty members gain AI-powered teaching tools for generating assignments, quizzes, and timed class tests, managing attendance, publishing materials, auto-grading assessments, providing AI-assisted feedback, and analyzing course performance.

Administrators gain centralized control over users, departments, academic structures, document approvals, permissions, announcements, and institution-wide analytics.

5. Feasibility

From a technical perspective, UniGPT is highly feasible and production-ready. The system is built using modern scalable technologies including Laravel 11, Vue.js 3, Inertia.js, Tailwind CSS, and MySQL. Its modular architecture supports secure role-based access control, efficient document indexing, AI provider abstraction, and maintainable long-term development.

The system has already implemented major admin, faculty, and student functionalities, demonstrating strong real-world applicability.

6. Scalability & Future Vision

UniGPT is designed with scalability in mind. Its architecture can support thousands of students, faculty members, departments, and documents across multiple institutions.

Future expansion includes real-time messaging, integration with messaging platforms like WhatsApp and Telegram, predictive analytics, voice-based AI interaction, digital library integration, and support for multiple AI providers.

The long-term vision is to transform universities into AI-driven intelligent ecosystems where academic knowledge becomes instantly accessible, administrative friction is minimized, and educational productivity is significantly enhanced.